

Laplace transform

1. Definition and existence

Definition 14.1 (Laplace transform). The Laplace transform of $f(t)$ is the improper integral

$$\mathcal{L}\{f(t)\} = \underline{\hspace{10em}} =: F(s),$$

whenever the integral converges. The variable s is complex:

$$s = \underline{\hspace{2em}}, \quad \sigma, \omega \in \mathbb{R}.$$

Compare with the Fourier transform.

The Fourier transform, a central object in ECE216, is

$$\mathcal{F}\{f(t)\}(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt.$$

Complete the two differences noted in the lecture:

- (a) In the Fourier transform, $\sigma = \underline{\hspace{2em}}$.
- (b) Its lower limit of integration is $\underline{\hspace{2em}}$, whereas the Laplace transform starts at $\underline{\hspace{2em}}$.

Theorem 14.2 (Existence of the Laplace transform). The integral defining $\mathcal{L}\{f(t)\}$ converges for sufficiently large $\text{Re}(s)$ provided that

- (a) f is $\underline{\hspace{10em}}$ on every finite interval in $[0, \infty)$;
- (b) f is of $\underline{\hspace{10em}}$.

We specify the resulting half-plane of convergence on the next page.

What does piecewise continuous mean?

On any finite interval $[a, b]$, f is continuous except possibly at a $\underline{\hspace{2em}}$ of discontinuities. At each discontinuity, the relevant one-sided limits exist and are $\underline{\hspace{2em}}$.

Sketch a piecewise continuous function with jump discontinuities.



2. Exponential order and the unit step

Definition 14.3 (Exponential order). The function f is of exponential order if there are constants $c \in \mathbb{R}$, $M > 0$, and $T \geq 0$ such that

$$|f(t)| \leq \text{_____} \quad \text{for all } t \geq \text{_____}.$$

Sketch the eventual exponential bound. Label T and Me^{ct} .



Complete the convergence conclusion of the existence theorem.

If f is piecewise continuous and satisfies the bound above, the Laplace integral converges for every s in the half-plane

$$\{s \in \mathbb{C} : \text{_____}\}.$$

The set of values of s for which the integral converges is called the _____.

Example 14.1 – The unit step function. Consider

$$u(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Check the hypotheses of the existence theorem.

The unit step is _____. For $t \geq 0$,

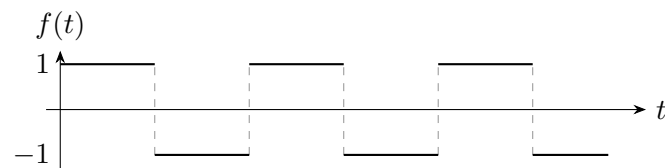
$$|u(t)| = 1 \leq 1e^{0t}, \quad M = \text{_____}, \quad c = \text{_____}, \quad T = \text{_____}.$$

Use the definition to compute $U(s) = \mathcal{L}\{u(t)\}$.

Evaluate the improper integral and explain the condition on s .

3. Further examples and a table of transforms

Example 14.2 – The square wave. Consider the square wave shown below, alternating between 1 and -1 .



Why is this function Laplace transformable?

Remark (A contrasting function). For $f(t) = 1/t$, which part of piecewise continuity fails at $t = 0$?

Table of transforms

Use the Laplace Transform definition to complete the table. Here $a, \omega \in \mathbb{R}$. Make sure to specify ROC.

Signal $f(t)$	Transform $F(s)$
$u(t)$	
$e^{at}u(t)$	
$\sin(\omega t)u(t)$	
$\cos(\omega t)u(t)$	